

YRGYZ, Kazakhstan

WMO: 355420

Lat: 48.617N

Lon: 61.267E

Elev: 374

StdP: 14.50

Time Zone: 5.00 (E05)

Period: 82-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-17.3	-13.9	-22.9	1.6	-16.6	-19.4	1.9	-13.5	36.7	8.9	34.1	19.3	7.4	340	0.623

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.2	96.9	64.8	93.3	64.2	90.0	63.4	67.7	87.4	66.4	87.0	65.2	85.3	9.0	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.9	83.9	73.2	59.9	78.0	72.5	57.6	71.8	72.6	32.3	86.8	31.2	87.6	30.3	85.0	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 30.3	26.3	22.1	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	-22.8	72.5	4.8	3.2	-26.3	74.8	-29.1	76.6	-31.8	78.4	-35.3	80.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.8	7.6	6.8	19.5	47.6	61.8	73.6	76.8	71.8	60.4	43.7	27.0	14.3
	DBStd	27.16	11.67	12.15	11.59	11.27	7.91	7.35	6.24	6.48	8.22	8.39	9.95	11.27
	HDD50	5693	1314	1209	947	178	7	0	0	0	12	229	691	1106
	HDD65	9091	1779	1629	1411	525	155	17	3	19	182	659	1141	1571
	CDD50	3056	0	0	0	107	374	707	831	677	325	35	0	0
	CDD65	979	0	0	0	4	57	274	368	231	45	0	0	0
	CDH74	11508	0	0	0	67	716	3358	4439	2427	497	4	0	0
	CDH80	5084	0	0	0	7	202	1603	2185	961	126	0	0	0
Wind	WSAvg	9.5	9.9	11.0	11.2	9.9	10.0	9.0	8.9	8.5	8.8	9.3	9.3	9.0
Precipitation	PrecAvg	6.9	0.4	0.4	0.5	0.6	0.7	0.8	0.7	0.5	0.5	0.6	0.6	0.6
	PrecMax	12.4	1.3	1.1	1.7	3.5	2.8	3.9	6.3	2.0	3.0	2.0	1.6	1.3
	PrecMin	3.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1
	PrecStd	2.0	0.3	0.3	0.4	0.7	0.7	0.7	1.1	0.5	0.6	0.5	0.4	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.3	34.1	52.4	79.9	88.6	101.8	102.8	96.9	88.2	73.2	47.3	36.1
		MCWB	31.7	32.5	44.4	57.0	59.7	65.8	66.4	64.1	59.1	54.2	40.8	34.0
	2%	DB	31.1	32.3	42.8	74.9	85.2	97.0	97.2	92.5	83.5	67.4	43.5	33.5
		MCWB	30.1	31.2	38.1	55.2	58.4	64.7	65.3	63.4	58.1	52.3	38.1	32.2
	5%	DB	29.0	29.9	37.8	70.4	81.6	92.4	94.0	89.0	80.1	62.4	40.9	32.0
		MCWB	28.2	28.8	34.6	53.9	57.4	63.9	64.8	62.7	57.0	49.0	37.3	31.1
	10%	DB	26.2	25.5	34.5	65.4	77.8	88.9	90.9	85.7	75.7	57.8	38.4	30.2
		MCWB	25.4	24.5	32.8	51.6	56.7	62.9	64.3	61.8	55.8	46.3	35.2	29.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.2	33.1	45.6	60.1	62.9	69.4	69.4	68.5	64.3	56.6	43.1	34.6
		MCDB	33.2	33.7	50.8	73.9	75.4	90.2	87.9	85.0	77.8	71.4	45.1	35.4
	2%	WB	30.2	31.2	38.4	57.3	61.3	67.4	68.0	66.0	61.5	53.3	39.6	32.7
		MCDB	30.8	32.1	42.2	70.3	77.4	87.6	88.0	83.5	75.5	64.2	42.0	33.4
	5%	WB	28.2	28.7	35.1	54.6	59.6	65.7	66.8	64.8	58.6	50.3	37.5	31.2
		MCDB	28.8	29.7	37.4	68.2	76.4	86.1	88.2	83.0	76.0	60.3	40.2	31.9
	10%	WB	25.5	24.4	32.7	52.1	58.0	64.5	65.8	63.5	56.7	47.3	35.3	29.2
		MCDB	26.3	25.5	34.4	64.7	74.3	85.0	87.3	81.2	74.3	56.6	38.0	30.2
Mean Daily Temperature Range	5% DB	MDBR	15.0	15.0	14.6	19.7	22.6	23.3	22.2	22.0	22.6	19.6	12.5	13.3
		MCDBR	17.7	13.0	14.0	25.9	27.3	28.0	26.3	25.7	27.7	26.7	15.3	11.1
		MCWBR	16.5	11.2	10.3	11.1	11.1	8.4	8.3	9.5	12.4	14.8	11.8	10.8
	5% WB	MCDBR	16.5	11.2	13.3	23.4	23.2	23.8	22.8	22.4	24.1	23.7	13.5	11.0
		MCWBR	15.9	11.0	10.0	10.9	10.8	8.7	8.9	9.8	12.0	14.4	11.4	10.7
Clear-Sky Solar Irradiance	taub	0.308	0.334	0.401	0.431	0.429	0.424	0.416	0.415	0.365	0.355	0.341	0.317	
	taud	2.094	2.023	2.015	2.065	2.151	2.226	2.266	2.247	2.339	2.279	2.213	2.128	
	Ebn at Noon	227	250	251	258	265	266	267	261	264	243	215	206	
	Edh at Noon	28	38	46	48	46	43	41	40	33	29	25	24	
All-Sky Solar Radiation	RadAvg	472	858	1215	1581	2048	2189	2097	1850	1426	850	421	302	
	RadStd	82	127	130	151	142	119	99	111	72	75	45	54	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page